CSA5112 - Cryptography and Network Security

3. Polyalphabetic (Vigenère) Cipher

Aim: To implement the Vigenère Cipher for encrypting plaintext using a keyword.

Algorithm:

1. Start.
2. Read plaintext and keyword.
3. Repeat the keyword until it matches the plaintext length.
4. Encrypt each letter using modular addition.
5. Display the ciphertext.
6. Stop.

Code:

```
#include<stdio.h>

#include<string.h>

int main()

{

    char text[100], key[100];

    int i, len;

    printf("Enter Plain Text: ");

    scanf("%s",text);

    printf("Enter Key: ");

    scanf("%s",key);

    len=strlen(key);

    printf("Cipher Text: ");

    for(i=0;text[i]!='\0';i++)
```

```
    {
        char c=((text[i]-'A')+(key[i%len]-'A'))%26+'A';
        printf("%c",c);
    }
    return 0;
}
```

Sample Input:

```
Plain Text: HELLO
Key: KEY
```

Sample Output:

```
Cipher Text: RIJVS
```

Result:

The Vigenère Cipher was successfully implemented, and the plaintext was encrypted using the specified keyword.